

Arya Shrestha

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EDUCATION

Marist University

Poughkeepsie, NY

Bachelor of Science in Computer Science CGPA: 3.9

Expected May 2028

- **Relevant Courseworks:** Data Structures & Algorithms, Object-Oriented Programming, Software Development I & II, Database Management (SQL), Discrete Mathematics

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, Typescript, SQL

Developer Tools: Git & GitHub, Docker, Kubernetes

Additional Skills: Agile & Scrum, SDLC, Unit, Integration & End-to-end Testing, NoSQL (MongoDB), AWS

EXPERIENCE

Freelance Software Engineer

September 2025 – January 2026

Legacy Louisiana

Covington, LA — Remote

- Developed full-stack estate planning platform for 60+ attorneys using Express.js, PostgreSQL, and Redis sessions, achieving 95% reduction in document generation time
- Implemented enterprise-grade authentication with bcrypt hashing, Redis session management, and multi-device tracking to secure attorney-client privileged documents
- Engineered scalable backend with versioned REST APIs and multi-tenant database architecture to support 16x growth from 60 to 1,000+ concurrent users

Software Engineer Intern

June 2025 – September 2025

Docere

St Peter, MN

- Developed a learning platform using React, Typescript, and Express serving 50+ users, achieving a 95% retention rate in initial deployments, projected to go live on Fall 2026.
- Engineered modular REST APIs for real-time progress tracking, directly securing \$9,000 in expansion funding from angel investors and private sources.
- Automated communication pipelines via SMS and email using Twilio and Nodemailer, increasing user engagement in underserved regions.

PROJECTS

Distributed Rate Limiter | Javascript, Redis, Nginx, Docker

Jan 2026

- Prevented API abuse across multiple Node.js servers by implementing token bucket rate limiting with Redis as shared state and Nginx load balancing (5 requests per 10-second window)
- Eliminated per-server request count bypass vulnerabilities by architecting distributed rate limiter using ioredis for centralized token management across stateless Express.js instances
- Achieved consistent rate enforcement under distributed load by designing Docker-containerized microservices architecture with round-robin load balancing and sub-second token refill granularity

Posturize | React JS, Flask, OpenCV, Mediapipe, D3.js

November 2025, **HackPrinceton**

- Awarded **Best Rookie Hack** at Princeton's largest hackathon by building real-time posture analysis system solo in 36-hour sprint
- Achieved sub-200ms posture feedback latency for 30+ landmark points by architecting Flask REST APIs with OpenCV and MediaPipe pose detection pipeline
- Increased user engagement through D3.js visualization dashboard displaying real-time shoulder alignment and spinal angle metrics with automatic posture scoring

LEADERSHIP & INVOLVEMENTS

HeadStart Fellowship

Spring 2026

Tech Fellow - Technical Interview Prep, F500 mentorship and direct recruiting connections

Propel2Excel

2025 - Present

Phase 2 Fellow (1% acceptance) - 1:1 mentorship and interview coaching

ColorStack

2025 - Present

Member - Conducted resume reviews and 1:1 CS advising for freshmen